BÀI TẬP VỀ NHÀ TUẦN 3 - PPS TRONG ĐSTT

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Bài tập 1.

Giả sử $A = UV^*$, trong đó

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \qquad \text{và} \qquad V = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Chứng minh, $\|A\|_2 = \|U\|_2.\|V\|_2$.

Lời Giải.

Ta có:

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \qquad \text{và} \qquad V^* = \begin{bmatrix} \bar{v}_1 & \bar{v}_2 & \dots & \bar{v}_n \end{bmatrix}$$

Suy ra, $A = U.V^* = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \cdot \begin{bmatrix} \bar{v}_1 & \bar{v}_2 & \dots & \bar{v}_n \end{bmatrix} = \begin{bmatrix} u_1\bar{v}_1 & u_2\bar{v}_1 & \dots & u_n\bar{v}_1 \\ u_1\bar{v}_2 & u_2\bar{v}_2 & \dots & u_n\bar{v}_2 \\ \vdots & \vdots & \ddots & \vdots \\ u_1\bar{v}_2 & u_2\bar{v}_2 & \dots & u_n\bar{v}_n \end{bmatrix}$.

Cho $\vec{x} = (x_1, x_2, \dots, x_n)$. Khi đó,

$$\begin{aligned} A\vec{x} &= \begin{bmatrix} u_1\bar{v}_1 & u_2\bar{v}_1 & \dots & u_n\bar{v}_1 \\ u_1\bar{v}_2 & u_2\bar{v}_2 & \dots & u_n\bar{v}_2 \\ \vdots & \vdots & \ddots & \vdots \\ u_1\bar{v}_2 & u_2\bar{v}_2 & \dots & u_n\bar{v}_n \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} \sum_{i=1}^n u_i\bar{v}_1x_i \\ \sum_{i=1}^n u_i\bar{v}_2x_i \\ \vdots \\ \sum_{i=1}^n u_i\bar{v}_nx_i \end{bmatrix} \end{aligned}$$

Suy ra,

$$
\begin{aligned}
\|A\|_2 &= \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} \frac{\|A\vec{x}\|_2}{\|\vec{x}\|_2} = \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} = \frac{\sqrt{\sum_{j=1}^n \left|\sum_{i=1}^n u_i\bar{v}_j x_i\right|^2}}{\|\vec{x}\|_2} \\
&= \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} \frac{\sqrt{\sum_{j=1}^n |\bar{v}_j|^2}.\left|\sum_{i=1}^n u_i x_i\right|}{\|\vec{x}\|_2} \leq \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} \frac{\sqrt{\sum_{j=1}^n |\bar{v}_j|^2}.\sum_{i=1}^n |u_i x_i|}{\|\vec{x}\|_2} \\
&\leq \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} \frac{\sqrt{\sum_{j=1}^n |v_j|^2}.\sqrt{\sum_{i=1}^n |u_i|^2}.\sqrt{\sum_{i=1}^n |x_i|^2}}{\|\vec{x}\|_2} = \sup_{\vec{x}\in\mathbb{C}^n, \vec{x}\neq\vec{0}} \frac{\|V\|_2.\|U\|_2.\|\vec{x}\|_2}{\|\vec{x}\|_2} \\
&= \|U\|_2.\|V\|_2.
\end{aligned}
$$

Dấu "=" xảy ra khi và chỉ khi $\|A\|_2 = \|U\|_2.\|V\|_2$.

Vậy, ta có điều phải chứng minh. $\square$